1D DP - Lecture 4

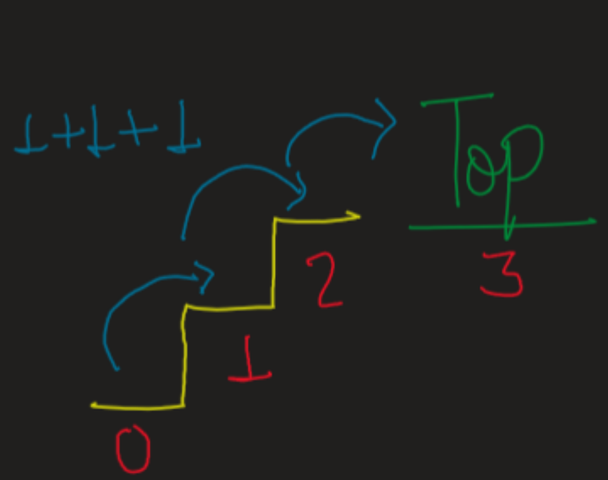
06 November 2025 17:30

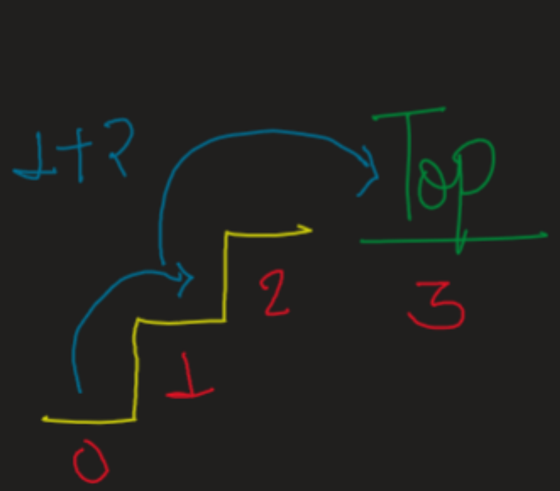
* Always Remember:→ Try Your Best to come up with the Recursive logic only, Because after that DP is just 5 minutes of work. – vHizen

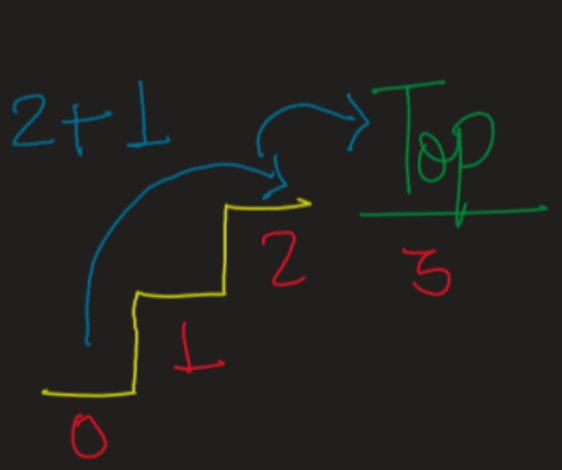
* Problem :-

You are climbing a staircase. It takes n steps to reach the top.
Each time you can either climb 1 or 2 steps. In how many distinct ways can you climb to the top?

Input: n = 3
Output: 3
Explanation: There are three ways to climb to the top.
1. 1 step + 1 step + 1 step
2. 1 step + 2 steps
3. 2 steps + 1 step







1 way + 1 way + 1 way = 3 ways

↝ go from 0 to 3

↝ go from 3 to 0

→ Both are same thing

1+1+1

1+2

2+1

1 way + 1 way + 1 way = 3 ways

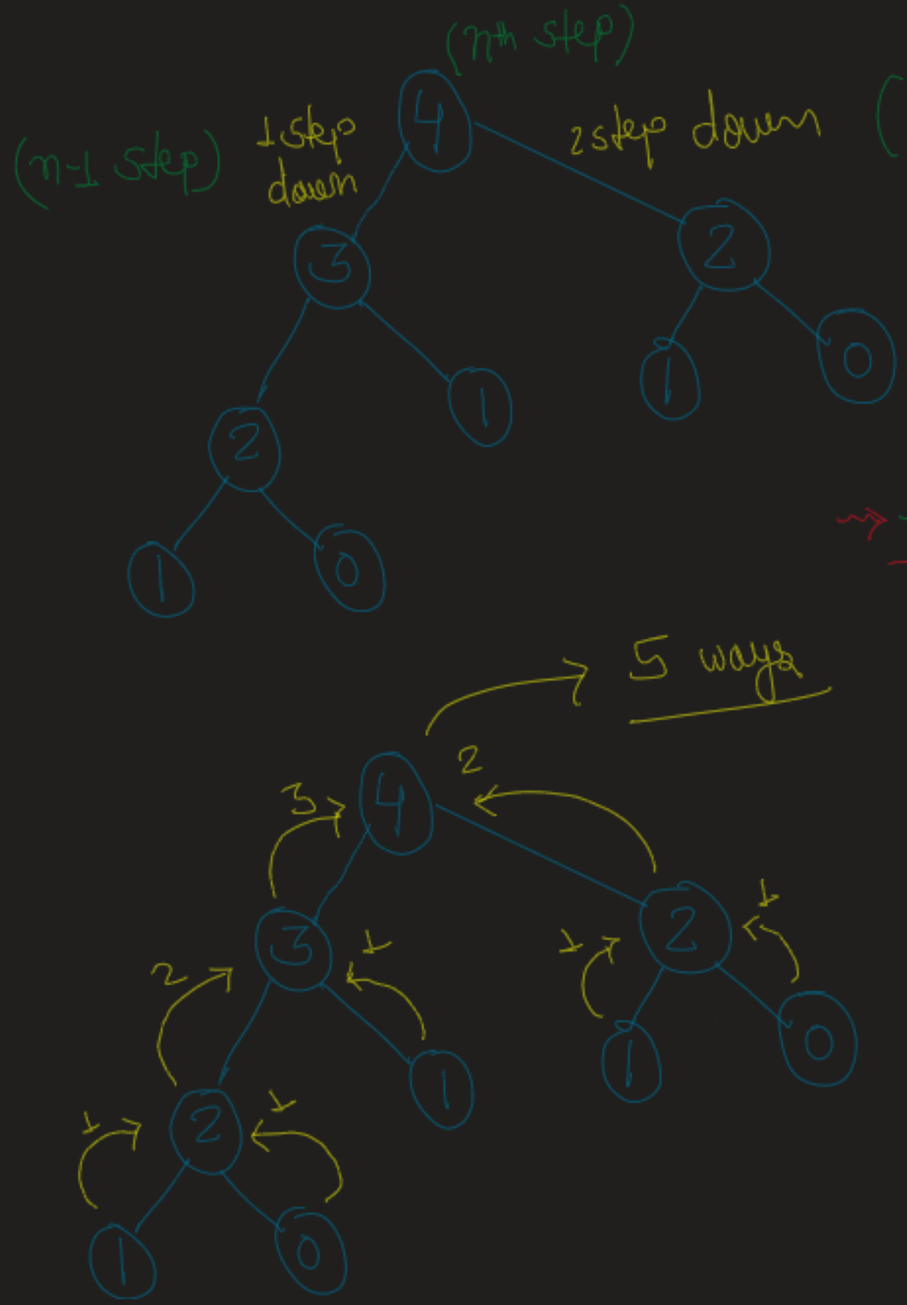
* How Question Provides Hints to think Recursive?

(i) Tries to give you options to explore

(ii) Tells you can't go greedy, must explore all possibilities/options to build result

* Lets Think Recursively :- (I am starting from n)

n = 4



→ find ways/count

↳ always add what you are returning

→ Totally related to fibonacci number

→ Just created a language in this

→ Wants you to find total ways to reach 0^{th} fib. number from n^{th} fib. number

* Time → $O(2^N)$, n = 4 | 4, 3, 2, 1

→ For any value we're exploring two options / possibilities.

: Why It's taking Exponential time?

→ Overlapping Sub problems (Repitative work)

→ Means when n is increases, the amount of sub-problem increases

→ Solving this repetative problems again and again is increasing time.

→ This concludes that we could optime time in our recursion.

→ Now this hints to DP.

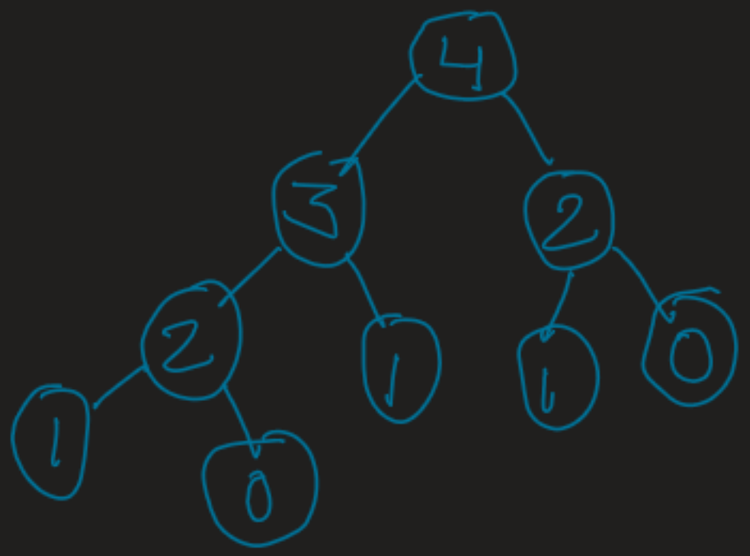
→ So for recursion, we apply Top-Down DP,

So lets do Memoization.

* let's identify Base cases :-

→ if (n == 1) return 1

→ if (n == 0) return 1



* For rest part :-

(we just need previous two counts):

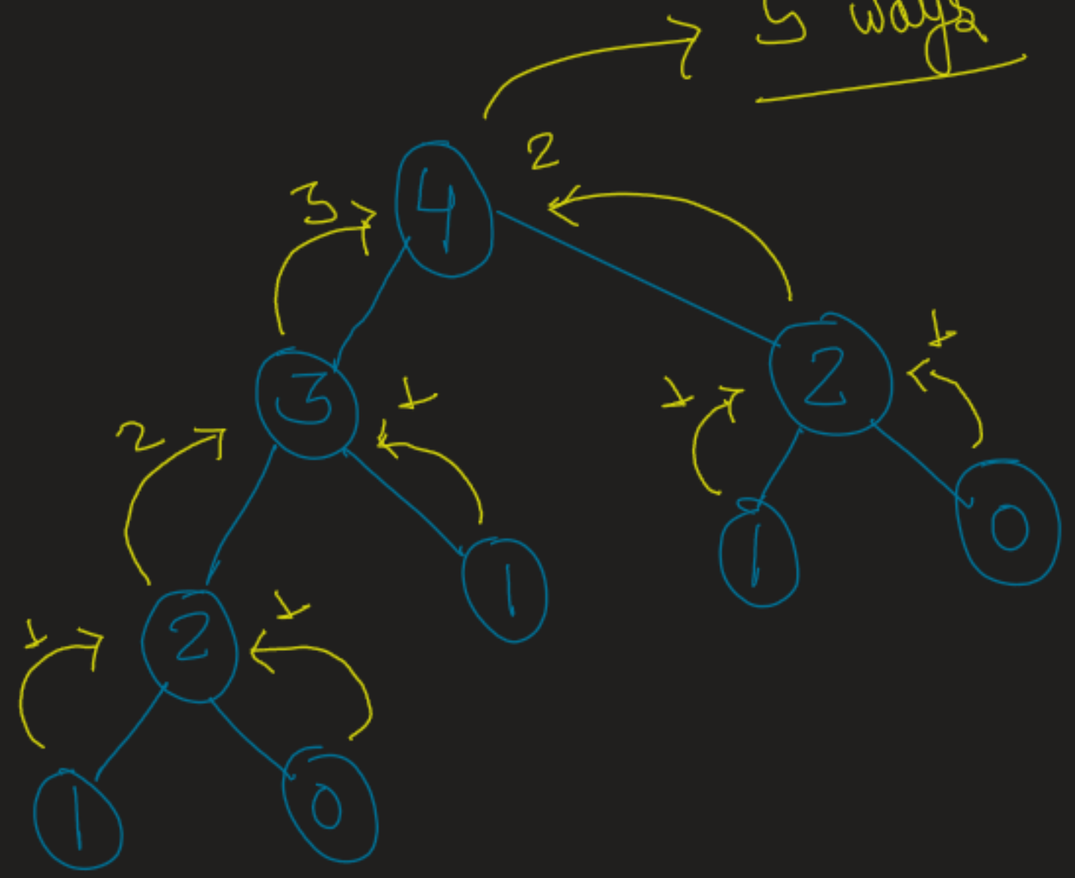
→ int count1 (find recursively)

int count2 (find recursively)

→ return sum of counts

~ Recursive code:-

```
int ways (int n){
    if (n==0) return 1;
    if (n==1) return 1;
    int JumpsStepDown = ways(n-1);
    int JumpsStepDown = ways(n-2);
    return JumpsStepDown + JumpsStepDown;
}
```



→ Now memoize it (Apply DP)

→ let's code it